

Research article

Variation in estimates of the footprint of large, ground-mounted photovoltaic solar energy in the United States and its associated land-cover change across three datasets

Michael O. Levin^{a,b,*}, Emma Forester^{b,c}, Elizabeth L. Kalies^d, Lauren Goodman^d, Jason S. Hagani^e, Chevon C. Holmes^c, Noah Z. Krasner^{b,c}, Caitlin Markus^f, Jared B. Meek^a, Anubhav Vanamamalai^a, Shivani Agarwal^a, Uzma Ashraf^{b,c}, Daphne Condon^{b,c}, Deborah Forester^g, Lucy C.V. Holland^a, Elizabeth Jackson^a, Madeline Levin^g, Patrick F. McKenzie^h, Benjamin P. Narwold^{b,c}, Linnea E. Norton^a, Breanna Rodgers^g, Lucy C. Cheadle^g, David B. Dayanⁱ, Julia Dayan^g, Andrew Levin^g, Rebecca R. Hernandez^{b,c}

^a Department of Ecology, Evolution, and Environmental Biology, Columbia University, New York, 10027, NY, United States

^b Wild Energy Center, University of California, Davis, Davis, 95616, CA, United States

^c Department of Land, Air, and Water Resources, University of California, Davis, Davis, 95616, CA, United States

^d The Nature Conservancy, Durham, 27701, NC, United States

^e School for Environment and Sustainability, University of Michigan, Ann Arbor, 48109, MI, United States

^f Nicholas School of the Environment, Duke University, Durham, 27708, NC, United States

^g Independent Researcher

^h Department of Organismic and Evolutionary Biology, The Arnold Arboretum, Harvard University, Cambridge, 02138, MA, United States

ⁱ Wildlife Health Lab, Animal Health Diagnostic Center, College of Veterinary Medicine, Cornell University, Ithaca, 14853, NY, United States

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ABSTRACT

Solar energy contributed more than six percent of the United States' (US) energy generation capacity in 2022, predominantly from large, ground-mounted photovoltaic solar facilities (GPVs). That proportion is expected to increase as GPV development accelerates and its land footprint expands. This study compared and harmonized three datasets describing the US GPV footprint, evaluated the land-use and land-cover change associated with each, and identified their best use cases. Two publicly available datasets delineated photovoltaic panel arrays, while a third paywalled dataset mapped the broader GPV fence line. For GPVs represented in all three datasets, we harmonized the data to 2018 and calculated both footprint area and quantified the footprint area and that of the land-cover classes converted within it. Our results suggest that US GPV footprint estimates from the panel array datasets are up to 34 % smaller than those from the fence line dataset. This has implications for land-cover change estimates; conversion of the relatively undisturbed categories shrubland and scrubland was 8 % higher in the fence line dataset than in the array datasets. As such, due to differences in their definitions of "footprint" and footprint delineation methods, these datasets are not interchangeable and have specific analytical contexts where they are best suited. A lack of publicly available data describing the broader GPV footprint hinders analysis of the relationship between GPVs and other place-based phenomena, particularly considerations of land-cover change and its relationship to biodiversity conservation.

* Corresponding author. Schermerhorn Extension 10th Floor, 1200 Amsterdam Ave, New York, 10027, NY, United States.

E-mail addresses: mol2112@columbia.edu (M.O. Levin), ekforester@ucdavis.edu (E. Forester), elizabeth.kalies@tnc.org (E.L. Kalies), lauren.goodman@tnc.org (L. Goodman), jhagani@umich.edu (J.S. Hagani), ccholmes@ucdavis.edu (C.C. Holmes), nzkrasner@ucdavis.edu (N.Z. Krasner), caitlin.m2015@gmail.com (C. Markus), jm4761@columbia.edu (J.B. Meek), vanubhav@gmail.com (A. Vanamamalai), shivani.2909@gmail.com (S. Agarwal), uashraf@ucdavis.edu (U. Ashraf), dcondon@ucdavis.edu (D. Condon), Debforester@gmail.com (D. Forester), lch2156@columbia.edu (L.C.V. Holland), ej2385@columbia.edu (E. Jackson), madeline.levin12@gmail.com (M. Levin), p.mckenzie@columbia.edu (P.F. McKenzie), bpnarwold@ucdavis.edu (B.P. Narwold), ln2489@columbia.edu (L.E. Norton), breaanna.rodgers97@gmail.com (B. Rodgers), lucycheadle@gmail.com (L.C. Cheadle), dbd63@cornell.edu (D.B. Dayan), juliadayan3@gmail.com (J. Dayan), drewlevin19@gmail.com (A. Levin), rrhernandez@ucdavis.edu (R.R. Hernandez).

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1. Introduction

Renewable energy generation technologies are a vital and burgeoning component of efforts to combat climate change. Driven by a patchwork of national and sub-national policy incentives (Aquila et al., 2017; Tryndina et al., 2022), global renewable energy capacity grew from 1851 GW (GW) in 2014 to 4448 GW in 2024 (IRENA, 2025). Among the most substantial contributors to this rapid growth is photovoltaic (PV) solar energy generation; the wide availability of sunlight as an energy generation resource and increasing economic competitiveness of PV technology has enabled PV solar to become the fastest growing energy source in the world, growing from an installed global capacity of 222 GW in 2014 to 1859 GW by 2024 (IRENA, 2025; Kabir et al., 2018; Nijse et al., 2023).

As PV solar capacity has expanded, so has its physical footprint. Among the most prominent global sources of electricity, large, non-residential, ground-mounted photovoltaic solar energy facilities (GPVs) have the third highest land-use intensity of energy (2000 ha/TWh/y) behind biomass (58,000 ha/TWh/y) and wind when accounting for spacing between turbines (12,000 ha/TWh/y (Lovering et al., 2022)). This may translate to substantial areas of land at the national level being devoted to PV solar energy generation; one analysis estimates that up to 1.4 % of land area in India, 2.8 % of that in the European Union, and 5.2 % of that in South Korea and Japan could be used for PV solar energy generation by 2050 (van de Ven et al., 2021).

Policies governing renewable energy development incentives in the United States (US) are mercurial. However, as of 2024, twenty-nine states and the District of Columbia adopted Renewable Portfolio Standards that require electricity suppliers in their jurisdiction to source increasing proportions of generation from renewable sources (Barbose, 2024). An estimated 73.5 GW (GW) of non-residential solar, six percent of the total US energy capacity, was operational by the end of 2022 (US Energy Information Administration, 2023); more than 2250 km² of land in the US hosted panel arrays by the third quarter of 2024 (Fujita et al., 2025). The capacity of operational US solar facilities increased by an annual average of 19.2 % between 2011 and 2021 (B.P. plc, 2022). In 2023, fifteen states hosted more than one GW of solar capacity from facilities with capacities of 5 MW or higher, including two, California and Texas, with over 15 GW (Seel et al., 2024). In three states—Nevada, California, and Utah—the percentage of in-state energy generation derived from solar installations larger than 1 MW exceeded 10 % in 2023, and in 13 others that percentage exceeded 5 % (Seel et al., 2024).

While there is a robust literature evaluating land for its solar energy generation potential (Hernandez et al., 2015; Oakleaf et al., 2019), far less attention has been devoted to digitizing developed solar energy infrastructure. Comprehensive data delineating the spatial extent of large-scale energy infrastructure have been scarce and/or difficult to access (e.g., privatized (Dunnnett et al., 2020)). Historically developed using manual and/or human-assisted delineation methods, datasets representing larger populations of facilities can be labor intensive to develop (Germaine et al., 2012). Despite advances in computing and the use of artificial intelligence, considerable human and computational effort is still needed to accurately digitize the unique, often complex polygon geometries of individual energy facilities across large spatial extents, including GPVs (Dai et al., 2024).

These hurdles to widespread development of rich, accessible spatial data describing energy infrastructure have led to the production of inconsistent, non-standardized datasets. In the wind energy context, for example, an assessment of onshore wind turbines in Sardinia, Italy found substantial discrepancies between three publicly available datasets, all of which underreported the number of onshore turbines in the region (Cerri et al., 2024). Similar issues have been documented for hydro-power and biomass databases and instigated similar efforts to rectify these shortcomings (Avitabile and Camia, 2018; Lin and Qi, 2019). Similar concerns apply to publicly-available data for solar energy, where spatially-explicit datasets of GPVs may identify the location of

individual facilities but record it using point data and other generalized geometries (e.g., U.S. Energy Information Administration, 2021). There is also a lack of standardization in defining the boundaries of a GPV, which means that the measured area or “footprint” of a facility is often unsuitable for comparison across studies (Cagle et al., 2022).

Solar energy expansion is inherently place-based and, as such, is likely to intersect with other spatial considerations in a given landscape. There is, for example, a small but growing body of literature describing the relationship between GPV infrastructure and biodiversity. While the infrastructural development process is intensive, the area within a GPV’s fenceline is not necessarily converted to an industrial, abiotic space. In human-mediated landscapes (e.g., agricultural regions) GPV development has been associated with increased species richness, diversity, and abundance, possibly due to local increases in suitable habitat within the GPV footprint (Cypher et al., 2019; Gerringer et al., 2022; Heredia-Velázquez et al., 2023; Walston et al., 2023). However, in relatively undisturbed landscapes (e.g. forests, deserts), GPV development is associated with primarily negative impacts on biodiversity, including detrimental alterations to microclimates and soil health, mortality and morbidity from collisions with facility infrastructure, and the alteration, fragmentation, and loss of habitat (Gasparatos et al., 2017; Gómez-Catasús et al., 2024; Grodsky et al., 2021; Grodsky and Hernandez, 2020; Kosciuch et al., 2020; Sawyer et al., 2022; Suuronen et al., 2017; Tanner et al., 2021; Walston et al., 2023).

There is thus an acute need for rich, accurate spatial datasets describing global and national GPV footprints to aid in assessing tradeoffs between GPV energy generation and other land-uses and land-covers. Advancements in machine learning and computer vision have facilitated a wave of data production that, in part, leverages the automatic detection of GPVs and rise in availability of high-resolution remote sensing imagery (Hu et al., 2022). In the US, several spatially-explicit datasets describing the array area of US GPVs—the relatively contiguous area that encompasses PV array infrastructure and the interstitial space between arrays—became publicly available in the early 2020’s. In addition, the US Photovoltaic Database, a solar-focused counterpart to the US Wind Turbine Database (Hoen et al., 2024), was released in 2023 and has since been updated several times (Fujita et al., 2023). However, GPVs can be delineated according to other criteria that may provide additional relevant information. Section 691.4 of the National Electric Code enforced in the US states that GPVs must have fences that 1) enclose the entirety of the facility, 2) reach at least seven feet tall, and 3) remain a minimum of 10 feet from electrical infrastructure, depending on voltage (National Fire Protection Association, 2017). Publicly available attempts to capture the US GPV footprint—the contiguous area within the fenceline, including the array footprint—are less common and typically comprise only a subset of the national GPV pool (e.g., California Energy Commission, 2023).

Here we present a comparison of datasets describing the spatial extent of US GPV array footprints and facility footprints. Our objectives are to (1) understand how the data collection methodologies and delineations of GPV “footprint” endemic to each dataset compare with one another; (2) compare temporally harmonized dataset-based estimates of the total GPV footprint across the US by the end of 2018 and 2021; and (3) compare temporally harmonized dataset-based estimates of GPV-driven land-cover change across the US by the end of 2018 and 2021. As data sharing and open science initiatives increasingly enhance data transparency and accessibility, it becomes equally important to understand the similarities, differences, and limitations of various datasets. Results from analyses of these datasets may differ and lack interchangeability, but still be used to drive decision- and policy-making (Bai and Fernandez, 2020; Rajagopalan et al., 2011; Subash et al., 2023). This analysis equips decision-makers with a comprehensive understanding of the magnitude and direction of land-related changes driven by GPV development, directing them toward the right data to capitalize on opportunities and navigate away from potential conflicts that may arise from the expansion of GPVs in the US.

2. Methods

2.1. Dataset acquisition and preparation

We assessed three spatial datasets that describe the area of operational GPVs to calculate the total area of GPVs in the US (Table 1). These particular datasets were chosen because, to the authors' knowledge, they are the only datasets that contain non-generalized, spatially-explicit data for US GPVs. Each dataset contains vector data in the form of polygons, closed shapes with geometries, or spatial extents, defined by a series of coordinate pairs that represent the footprint of individual GPVs. The first dataset, published in 2021 by Kruitwagen et al. (henceforth the Kruitwagen dataset), was developed using an automated machine learning pipeline to identify the array area of global PV infrastructure visible in remote sensing imagery as of 2018 [41] (Fig. 1). The authors trained convolutional neural networks and recurrent neural networks to analyze Sentinel-2 and SPOT-6/7 satellite imagery for the presence of PV arrays, vectorize their boundaries, remove potential false positives, and estimate the array's installation date. The authors reported that the intersection over union of their "... final dataset for installations over 10,000 m² is 90 %—sufficient for the wide range of uses that our global dataset enables," and that this accuracy was comparable to other automated PV solar remote sensing studies (Kruitwagen et al., 2021).

We spatially filtered the Kruitwagen dataset to include GPVs located within the US, including those within the states of Hawaii and Alaska. The number of polygons in the Kruitwagen dataset representing the array footprint of GPVs in the US through 2018 is 7680. Next, we assessed all polygons in this US subset to determine whether they accurately identified GPVs. The dataset was separated into bins of 100 polygons and distributed to a group of 25 validators, where each validator reviewed a minimum of six bins. Each bin was reviewed twice by different validators, and no validator pairs were repeated across bins. Validators uploaded their bins into Google Earth Pro (vers. 7.June 3, 9345) and used high-resolution imagery from the past and present to manually review each polygon and determine whether the dataset correctly captured and identified a given location as a GPV. We removed polygons that incorrectly represented a given location as a GPV ($n = 385$, 5 %). Those facilities where paired validators agreed that a polygon did not represent a GPV were removed, while those with divergent assessments between validators were removed only after review from a third validator.

The second dataset was the US Large-Scale Solar Photovoltaic Database (USPVDB), produced by the Lawrence Berkeley National Laboratory and United States Geological Survey. We used the first version of this dataset (v1_0_20231108) released in 2023. This US-based dataset utilized data from the US Energy Information Administration to identify the location of GPVs and used high-resolution aerial imagery to

delineate their array footprints (Fujita et al., 2023, Fig. 1). The number of polygons in the USPVDB representing the array footprints of GPVs in the US through 2021 is 3699. To ensure temporal continuity, we compared GPV attributes in the USPVDB with those in the Kruitwagen dataset. Whereas the Kruitwagen dataset contained an attribute describing the "installation date" of each solar facility, derived from the year of the satellite imagery where the facility could be visualized, the USPVDB contained an attribute entitled "p_year" describing the year a facility became operational, derived from a form that GPV developers must submit to the US Energy Information Administration. Based on the p_year attribute, we removed GPVs constructed after 2018. We did not independently validate the USPVDB, given the dataset's derivation from manual array delineations and high reported accuracy; the authors qualified their digitizing accuracy as "highly confident" or "confident" for 96 % of the USPBVD).

The third dataset is not publicly available and was purchased from the private company Wiki-Solar (<https://wiki-solar.org/>; henceforth the Wiki-Solar dataset). Wiki-Solar compiles and maintains a global database of solar facilities, including ground-mounted and rooftop. The polygons in this dataset purportedly delineate a GPV's fence line when possible, providing a spatial estimate of the facility footprint (Fig. 1). According to Wiki-Solar's published eligibility criteria (<https://wiki-solar.org/data/criteria.html>), this dataset includes GPVs with capacities greater than or equal to 3.5 MW (MW) and some facilities below that threshold when they are "noteworthy" or present on site plans associated with larger GPVs. Polygons were digitized using imagery from Google Maps; for GPVs, "the site area shown is all that is enclosed within any perimeter fencing for the project. Unfenced projects show the area of the solar arrays plus a surrounding border, to allow for the area are [sic] likely to be required for maintenance access. Widely dispersed sites - such as hilltop arrays - may show an irregular perimeter to enclose the area visibly covered by solar arrays ... For simplicity, wavy or crooked boundary lines may be shown as straight lines on our maps. To avoid complexity in some cases where roads, paths, streams, power lines and other features cross solar sites, these may be included in the indicated site area, even when they are fenced separately." Finally, Wiki-Solar limits the maximum number of coordinate pairs that can be used to define a polygon's geometry in their datasets to approximately 75 pairs. Where more pairs are needed to represent the spatial extent of a GPV with perfect accuracy, this restriction may generalize a GPV's spatial extent. However, it also prevents the geometries from becoming too complex, ensuring that all polygons can be displayed seamlessly and without lags on commonly used mapping platforms (e.g., Google Maps).

We spatially filtered the Wiki-Solar dataset to include only GPVs within the boundary of the US. Next, we assessed all polygons in the Wiki-Solar dataset to identify those constructed on rooftops or over parking lots and removed them from our analysis, as well as facilities with heliostats (i.e., infrastructure used for concentrating solar power).

Table 1

Characteristics of spatial datasets describing the footprint of large, ground-mounted photovoltaic solar energy facilities in the United States. Note that the temporal range of these datasets may be updated by more recent releases.

Dataset	Temporal Range	Method of Footprint Delineation	Scale	Accessibility	Citation or Website
Kruitwagen	Facilities constructed before or during 2018	Automatic Digitization (Machine Learning)	Global	Data is publicly available and open access	Kruitwagen, L., Story, K. T., Friedrich, J., Byers, L., Skillman, S., & Hepburn, C. (2021). A global inventory of photovoltaic solar energy generating units. <i>Nature</i> , 598(7882), 604–610.
US Photovoltaic Database	Facilities operational before or during 2021	Manual Digitization	The United States	Data is publicly available and open access	Fujita, K.S., Ancona, Z.H., Kramer, L.A., Straka, M., Gautreau, T. E., Garrity, C.P., Robson, D., Diffendorfer, J.E., and Hoen, B., 2023, United States Large-Scale Solar Photovoltaic Database (v1.0, November 2023): U.S. Geological Survey and Lawrence Berkeley National Laboratory data release, https://doi.org/10.5066/P9IA3TUS .
Wiki-Solar	Facilities operational before or up to 2025	Manual Digitization	Global	Data is privately owned and maintained behind a paywall	https://wiki-solar.org/services/

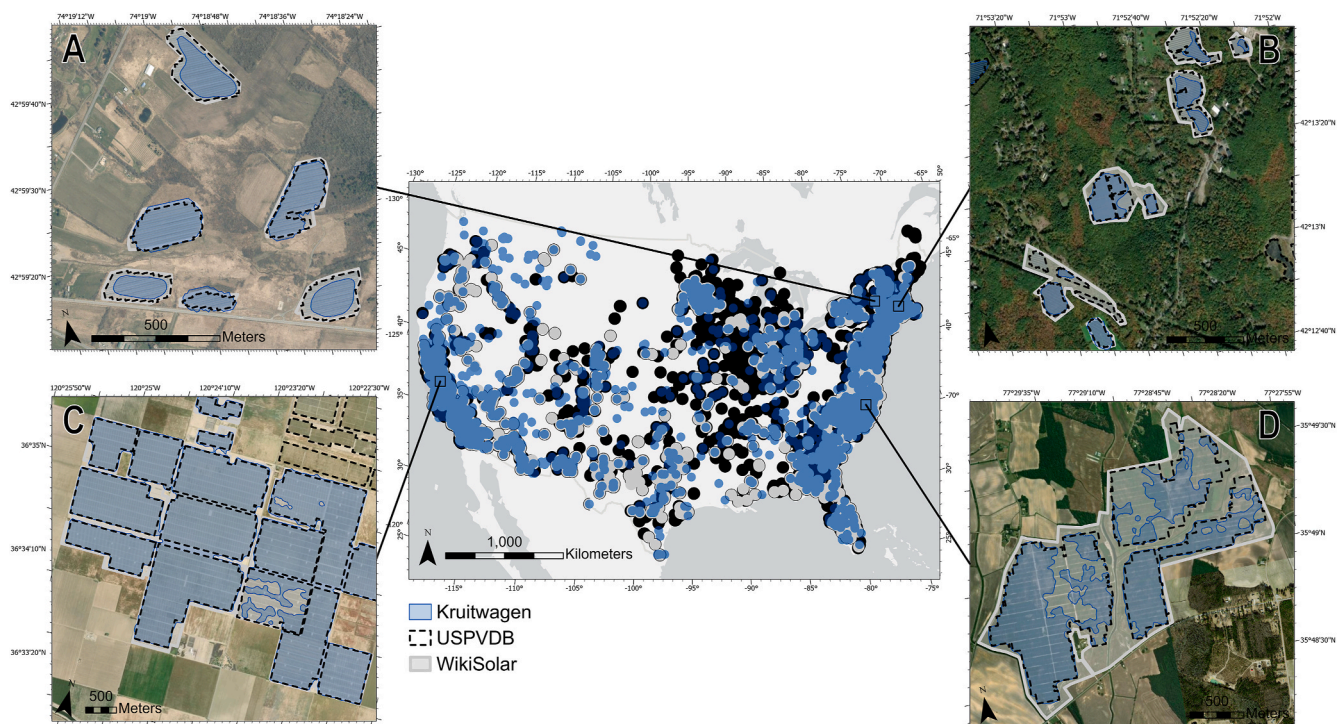


Fig. 1. We assessed three spatial datasets that describe the area of operational large, ground-mounted photovoltaic solar facilities (GPVs) to calculate the area of GPVs in the US. The Kruitwagen dataset (in blue) used an automated machine learning pipeline to identify global solar facilities visible in remote sensing imagery by the end of 2018 and delineate the area of photovoltaic panel arrays. The United States Photovoltaic Database (in black, with dashed borders) identified locations of solar facilities operational through the end of 2021, manually verified their presence with high-resolution aerial imagery, and delineated the area of photovoltaic panel arrays. The Wiki-Solar Dataset (in gray), a private dataset purchased from the company Wiki-Solar, identifies the location of solar facilities from several sources, manually verified their presence with high-resolution aerial imagery, and delineated the area of GPVs within their fencelines.

Furthermore, one of the attributes present in the Wiki-Solar dataset, ‘Status’, refers to the operating status of each facility. We removed polygons with the values ‘Stalled’, ‘Planned’, ‘Include former’ (i.e., decommissioned GPVs), and ‘Building’ (i.e., GPVs under construction) for that attribute as this indicated that the polygon does not represent an operational GPV. Facilities present in this dataset constructed after 2021 were removed from our analysis to align with the temporal range of the USPVDB and Kruitwagen datasets. The number of polygons representing the facility footprint of GPVs in the US through 2021 was 3846. To ensure temporal continuity, we compared GPV attributes in the USPVDB with those in the Kruitwagen dataset and removed GPVs constructed after 2018. The final number of polygons in the Wiki-Solar dataset through 2018 was 2301.

The Kruitwagen dataset records the first year a facility’s construction was visible in satellite imagery. The USPVDB and Wiki-Solar datasets record the year a facility became operational. Given that construction must take place before a facility becomes operational, this discrepancy in temporal data introduces the potential for overestimation of installed capacity from the Kruitwagen dataset when compared to the others assessed here. For example, a facility that began construction in 2017 but did not become operational until 2019 may be represented in the Kruitwagen dataset but not in the USPVDB nor Wiki-Solar dataset. Where facilities were represented in all three datasets, we found 140 Kruitwagen polygons representing an area of 38.9 km² with year data earlier than their USPVDB and Wiki-Solar counterparts. We also noted that 125 polygons representing an area of 19.1 km² had year attributes later than both their USPVDB and Wiki-Solar counterparts. While theoretically this could lead to underestimation of the area of the US solar facilities, this is less of a concern for this analysis as this type of discrepancy would not cause facilities to be excluded from the temporal

subsets we examine. Readers should thus be aware of the potential overestimation, but know the fraction of total facilities and area affected by these temporal mismatches is modest and suggests our results are robust to this potential uncertainty.

2.2. Accuracy assessment of the Wiki-Solar Dataset

We assessed the accuracy of a subset of the Wiki-Solar dataset by comparing the boundaries of Wiki-Solar polygons to any fenceline visible in high-resolution imagery present in Google Earth Pro. We assessed all Wiki-Solar polygons that overlapped with polygons in the Kruitwagen dataset to better confirm that a direct comparison between individual GPVs present in the two datasets would be valid.

We ranked Wiki-Solar polygons on a scale of 1 (Very Inaccurate) to 5 (Very Accurate). Validators assigned a 1 to polygons with little to no adherence to a facility’s fenceline, and 5 to those facilities where the polygon boundary adhered almost perfectly or perfectly to a facility’s fenceline (Supplementary Material A). Given Wiki-Solar’s documented polygon digitization methodology for unfenced facilities encompassing any panel arrays plus an unspecified “surrounding border,” we instructed validators to assign accuracy ratings based on general adherence to the panel area.

To assess interrater reliability among polygon validators, we calculated a weighted Cohen’s kappa and percentage agreement statistic. The kappa coefficient was 0.254, indicating fair agreement between validators (Cohen, 1960, 1968; McHugh, 2012). While a fair reliability coefficient suggests that agreement between validators is greater than what would be expected by chance, it may still fall short of an acceptable threshold. However, Cohen’s Kappa can be influenced by imbalanced rating distributions, sometimes yielding low Kappa values despite high

absolute agreement. This phenomenon is known as the Kappa Paradox (Derksen et al., 2024; Dettori and Norvell, 2020). In our dataset, overall agreement was high: 67 % ($n = 1269$) of the polygons received identical rankings from both validators. Approximately 25 % ($n = 470$) differed by one point (e.g., 4 and 5), while only 8 % ($n = 160$) had rankings that differed by greater than one point. A high volume of the rankings with perfect agreement (62 % of the total dataset) fell within the high accuracy category (5 - Very Accurate). These levels of absolute or near agreement are especially notable given the highly subjective and complex nature of the task, and the fact that 24 validators were randomly assigned from a pool to validate portions of Wiki-Solar dataset. To further support the credibility of our findings and account for potential misrepresentation caused by uneven data distribution, we also calculated Gwet's AC1, a measurement robust against the Kappa Paradox (Gwet, 2008). The weighted Gwet's AC1 coefficient was approximately 0.92, indicating near perfect agreement when unaffected by data imbalance.

The final accuracy score of polygons with rankings that differed by only a single point became the average of its two scores. Rankings that differed by more than one point were reviewed by a third validator. The third validator reviewed these polygons to investigate discrepancies between the two initial rankings and produce a third validation ranking, which was used as the final validation ranking for that polygon. We ultimately validated 1899 polygons of the 3846 in the US Wiki-Solar dataset that were operational by 2021, representing 49 % of the dataset. We scored 62 % ($n = 1172$) of validated Wiki-Solar polygons as 5 (Very Accurate), 21 % ($n = 397$) as 4.5, 9 % ($n = 164$) as 4 (Moderately Accurate), 4 % ($n = 83$) as 3.5, and 4 % ($n = 83$) as 3 (Neither Moderately Accurate nor Moderately Inaccurate) or lower. This means that nearly a third ($n = 1,172$, 31 %) of the US Wiki-Solar dataset operational by 2021 received the highest possible accuracy ranking.

2.3. GPV area calculation and comparison across datasets

We imported the harmonized Kruitwagen, USPVDB, and Wiki-Solar datasets into Google Earth Engine (Gorelick et al., 2017) and calculated the area of every polygon in each one. Next, we calculated the GPV footprint in each state and across the US as a whole by summing the areas of all the polygons in each state and summing these 50 values, respectively. Next, we spatially filtered each dataset to include only those polygons representing solar facilities that are present in the other two datasets. In addition, we filtered each individual dataset to include only those polygons that are spatially unique to that dataset (e.g., those polygons in the Kruitwagen dataset with no geographic intersections with polygons in either the USPVDB or the Wiki-Solar dataset).

Next, we accessed the 2004, 2006, 2008, 2011, 2013, 2016, 2019, and 2021 National Land Cover Database (NLCD) land-cover map products, all of which are publicly available, that describe 16 land-cover categories in the US at 30-m resolution (Dewitz, 2023). Using a grouped reducer in Google Earth Engine, we calculated the area of each land-cover class present in the NLCD found within the boundaries of GPV polygons from each dataset. Partial pixel overlaps were handled using Google Earth Engine's default mask-based weighting. Edge pixels with partial overlaps of solar facility polygons in each dataset were assigned fractional mask values, and the grouped reducer applied weighted reductions where each pixel's contribution was scaled by its mask value. As such, pixels with only partial overlap of solar facility polygons contributed proportionally to the area of the land-cover they represented.

All three GPV datasets contain attribute information indicating the year in which the GPVs became operational. This was used to link polygons to a particular NLCD dataset from which land-cover change calculations would then take place. For example, land-cover change calculations for facilities that became operational in 2004 and 2005

were calculated from the 2004 NLCD land-cover product, while GPVs that became operational in 2013, 2014, and 2015 would use the 2013 NLCD land-cover product. Land-cover classes were aggregated into broad categories. The four land-cover classes denoting various intensities of development were combined into a single 'developed' category. The three forest land-cover classes (deciduous, evergreen, and mixed) were combined into a single 'forest' category. Cultivated crops and pasture/hay were combined into a single 'crop and pastureland' category. Woody wetlands and emergent herbaceous wetlands were combined into a single 'wetlands' category. Here, we report results for all terrestrial land-cover categories, excluding the open water category.

3. Results

3.1. Kruitwagen dataset area and land-cover change estimates

For GPVs constructed by the end of 2018, the Kruitwagen dataset estimated a national array footprint of 623 km². Of that, 33 % (206 km²) was developed on crop and pastureland, 26 % (206 km²) on developed land, and 25 % (152 km²) on shrubland/scrubland (Figs. 2 and 3). Grassland/herbaceous land, forest, and barren land accounted for a combined 15 % (91 km²) of land developed for GPVs through 2018.

The Kruitwagen dataset contained 4039 unique polygons (55 % of the full dataset) that do not overlap with polygons in either the USPVDB or Wiki-Solar dataset, representing an area of 80 km² (13 % of the dataset's national footprint). Nearly all these polygons ($n = 4,028$, 99.7 %) have estimated footprints below 0.5 km². The entirety of the Kruitwagen dataset's estimated array footprint in Louisiana, West Virginia, and Washington can be found within the polygons that are unique to the Kruitwagen dataset, as can more than 50 % of the dataset's estimated array footprint in Iowa and New Hampshire.

3.2. USPVDB area and land-cover change estimates

For GPVs constructed by the end of 2018, the USPVDB estimated an array footprint of 668 km², 45 km² larger than that of the Kruitwagen dataset. Of that, 38 % (256 km²) of the GPV array footprint was developed on crop and pastureland, 24 % (161 km²) on developed land, and 22 % (148 km²) on shrubland/scrubland (Figs. 2 and 3). Grassland/herbaceous land, forest, and barren land accounted for a combined 14 % (94 km²) of land developed for GPVs through 2018. For facilities constructed by the end of 2021, the USPVDB estimated an array footprint of 1284 km². Of that, 39 % (499 km²) was developed on crop and pastureland, 26 % (338 km²) on shrubland/scrubland, and 17 % (215 km²) on developed land (Figs. 2 and 3). Grassland/herbaceous land, forest, and barren land accounted for a combined 17 % (215 km²) of land developed for GPVs through 2021. Note that the apparent numerical coincidence where the area of development on shrub/scrubland equals that of grassland/herbaceous land, forest, and barren land combined is due to rounding, and is not a data error.

The USPVDB contained 906 polygons (25 % of the full dataset) that do not overlap with polygons in either the Kruitwagen or Wiki-Solar datasets, representing an area of 169 km² (13 % of the dataset's national footprint). The entirety of the USPVDB's estimated array footprint in New Hampshire can be found within the polygons that are unique to the USPVDB, as can more than 50 % of the dataset's estimated array footprint in Illinois, Maine, Wisconsin, and Alabama.

3.3. Wiki-Solar Dataset area and land-cover change estimates

For GPVs constructed by the end of 2018, the Wiki-Solar dataset estimated a facility footprint of 939 km², 271 km² larger than that of the USPVDB. Of that facility footprint, 35 % (326 km²) was developed on crop and pastureland, 27 % (250 km²) on shrubland/scrubland, and 20

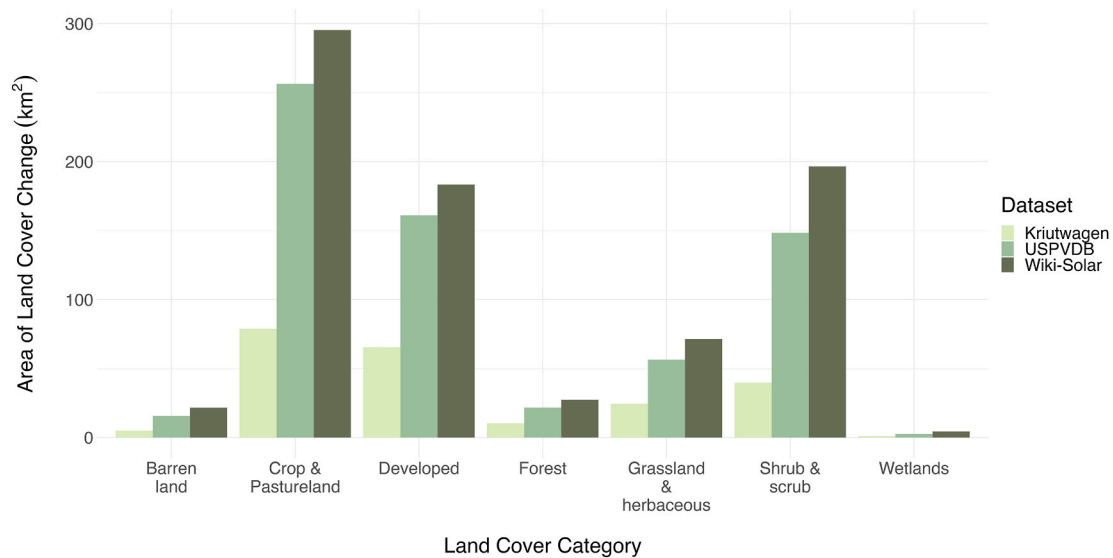


Fig. 2. Cumulative area (km²) of land-cover change associated with large, ground-mounted solar energy facilities (GPVs) in the United States (US) through 2018 across three geospatial datasets. The Kruitwagen dataset used a computer vision algorithm to delineate the array footprint of US GPVs, or the area shaded by photovoltaic panels and the interspaces between adjacent arrays. The United States Photovoltaic Database, or USPVB, relied on manual delineation to generate a dataset of the array area of US GPVs. The Wiki-Solar dataset also relied on manual delineation of US GPVs but delineated the full facility footprint—the array footprint in addition to the other uncovered areas of the PV solar project to the fenceline.

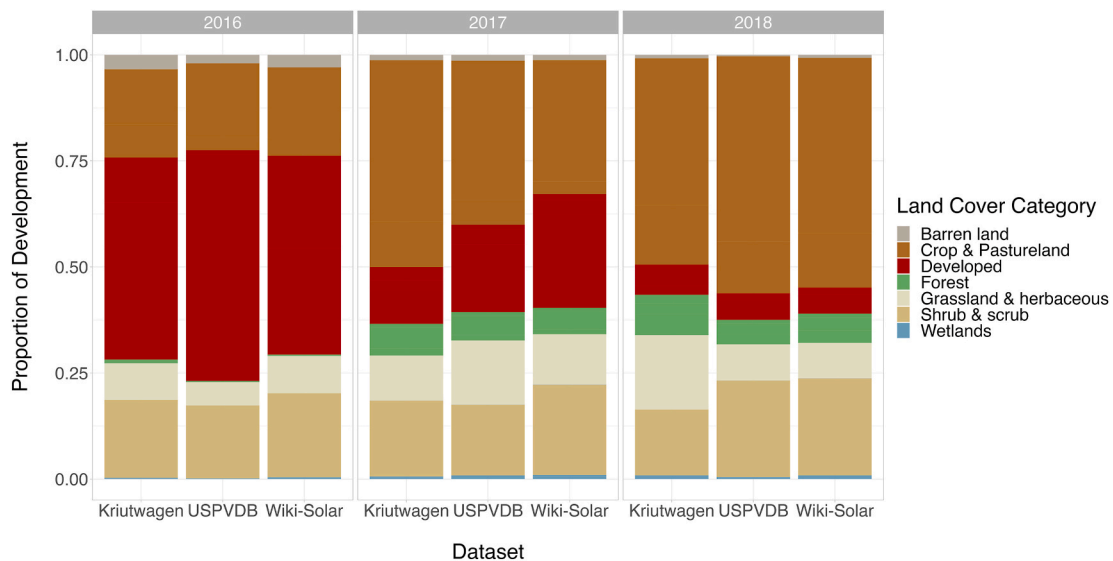


Fig. 3. Proportion of large, ground-mounted solar facility development in the United States within various land-cover categories in 2016, 2017, and 2018. The Kruitwagen dataset used a computer vision algorithm to delineate the array footprint of US GPVs, or the area shaded by photovoltaic panels and the interspaces between adjacent arrays. The United States Photovoltaic Database, or USPVB, relied on manual delineation to generate a dataset of the array area of US GPVs. The Wiki-Solar dataset also relied on manual delineation of US GPVs but delineated the full facility footprint—the array footprint in addition to the other uncovered areas of the PV solar project to the fenceline. The Kruitwagen dataset only has annual temporal attributes for these three years; all data prior to 2016 is aggregated into a single temporal category. As such, these three years are the only avenue for annual comparison across the three datasets. Total areas by dataset and year were 110, 75, and 40 km² for Kruitwagen; 190, 109, and 105 km² for USPVB; and 227, 137, and 133 km² for Wiki-Solar in 2016, 2017, and 2018, respectively. Note that the apparent numerical coincidence where the area of development on shrub/scrubland equals that of grassland/herbaceous land, forest, and barren land combined is due to rounding, and is not a data error.

% (192 km²) on developed land (Figs. 2 and 3). Grassland/herbaceous land, forest, and barren land accounted for a combined 17 % (161 km²) of land developed for GPVs through 2018. For GPVs constructed by the end of 2021, the Wiki-Solar dataset estimated a footprint of 1553 km², 269 km² larger than that of the USPVB. Of that, 39 % (610 km²) was developed on crop and pastureland, 25 % (384 km²) on shrub/scrub, and 16 % (248 km²) on developed land (Figs. 2 and 3). Grassland/herbaceous land, forest, and barren land accounted for a combined 18 %

(287 km²) of land developed for GPVs through 2021.

The Wiki-Solar dataset contained 278 polygons (8 % of the full dataset) that do not overlap with polygons in either the Kruitwagen datasets or USPVB, representing an area of 59 km² (4 % of the dataset's national footprint). Most polygons unique to the Wiki-Solar dataset (51 %) represented GPVs with capacities between 1 and 5 MW. The polygons that were unique to the Wiki-Solar dataset contained no more than 40 % of any state's estimated solar footprint, featuring 39 % of Rhode Island's

and 33 % of Maine's estimated solar footprint.

3.4. GPV and state area comparisons across datasets

The total area of polygons in the Kruitwagen dataset that intersected with at least one polygon in both the USPVDB and Wiki-Solar datasets ($n = 2,193$, 29 % of the US dataset) was 492 km². The total area of such polygons from the USPVDB ($n = 1,305$, 54 % of the 2018 subset) was 522 km². The total area of such polygons from the Wiki-Solar dataset ($n = 1,735$, 75 % of the 2018 subset) was 773 km². Overlapping polygons in the Kruitwagen dataset and the USPVDB ostensibly described the same panel array. The average difference between overlapping polygons from these datasets was 0.4 km². Of the area estimates for the 3265 polygons that intersect between these datasets, 82 % ($n = 2673$) differed by no more than 0.5 km² and 66 % ($n = 2144$) differed by no more than 0.1 km². Overlapping polygons in the Wiki-Solar dataset and the USPVDB described the same solar facility, but the Wiki-Solar dataset included area beyond that of the panel arrays. The average difference between overlapping Wiki-Solar and USPVDB polygons was 0.8 km². Of the area estimates for the 3051 polygons that intersected between these datasets, 75 % ($n = 2301$) differed by no more than 0.5 km² and 55 % ($n = 1671$) differed by no more than 0.1 km².

There were data on facilities constructed by the end of 2018 for all states in at least one of the three datasets. The Kruitwagen dataset lacked data for Wyoming and North Dakota; the USPVDB lacked data for North Dakota, Louisiana, New Hampshire, and West Virginia; the Wiki-Solar dataset lacked data for Kansas, Louisiana, New Hampshire, South Dakota, Wisconsin, and West Virginia. Where the USPVDB and Kruitwagen datasets have estimates for state GPV footprints, the USPVDB has higher area estimates for 26 states (including those of Texas, North Carolina, and Florida with a difference greater than 10 km²). The Kruitwagen dataset has higher area estimates for 18 states, including California, but by no more than 4 km². The average difference between the USPVDB and Kruitwagen dataset's estimated state solar footprints was 2 km². Between the USPVDB and Wiki-Solar Dataset, the average difference between estimated state solar footprints was 7 km². The USPVDB described larger solar footprints for 14 states compared to the Wiki-Solar dataset; no estimate was more than 4 km² above that of the Wiki-Solar dataset, and only two (Maine and New York) were more than 1 km² greater than that of the Wiki-Solar dataset. The Wiki-Solar dataset estimated larger state solar footprints for 28 states when compared to the USPVDB; estimates for Arizona, Nevada, Texas, and North Carolina were all over 10 km² greater than those of the Wiki-Solar dataset, and the estimate for California was 187 km² greater.

4. Discussion

Comparative analyses of renewable energy datasets have been undertaken globally for wind, hydropower, and biomass, highlighting the importance of data harmonization for a better understanding of land-use implications across renewable sources of energy. Our study extends this line of inquiry by focusing specifically on the US GPV footprint, where differences in how datasets define facility boundaries may have consequences for biodiversity conservation and land-use planning. Estimates of the total land area in the US developed for GPVs in the USPVDB and Kruitwagen datasets, which characterize only the array footprint, are respectively 29 % and 34 % smaller than that of the national facility footprint described by the Wiki-Solar dataset, which includes the broader facility footprint extending to the fenceline. Where polygons from all three datasets overlapped, the combined area of those facilities was over 200 km² greater in the Wiki-Solar dataset than in the USPVDB or Kruitwagen datasets. The differences between these datasets may at first appear semantic. However, our analysis of land-cover change driven by GPV development underscores how vital it is to understand

their differences. The criteria used to delineate a GPV's footprint are especially relevant when examining the relationship between GPVs and other place-based phenomena—particularly biodiversity conservation.

Given that habitat loss and alteration is the largest driver of global declines in biodiversity (Caro et al., 2022; Cousins et al., 2015; Jaur-eguiberry et al., 2022), accounting for changes in land-cover categories that are relatively undeveloped is necessary in the context of GPV development. The proportion of GPV development that occurred across land-cover categories is largely consistent between datasets, despite differences in cumulative footprint area. One exception, however, is scrubland/shrubland, a category that represents a relatively unaltered land-cover. In the Wiki-Solar dataset, 25 % of GPV development through the end of 2021 occurred on scrubland/shrubland, compared to 17 % in the USPVDB. Data describing a GPV's footprint provides a more accurate representation of real-world habitat loss and alteration and captures a driver of land-cover change that is more consequential for biodiversity than is represented in the array footprint alone.

It is worth noting, however, that land within the GPV footprint is not converted exclusively to energy generation. There is a growing consensus that land within the GPV footprint, and even land within the array footprint, can contribute to dual land-uses that support various levels of ecosystem functionality. An agrivoltaic GPV approach co-locates agricultural operations with PV infrastructure, providing some ecological functionality and additional avenues of revenue generation (Amaducci et al., 2018). Panels provide shade and reduce evapotranspiration, creating microclimates that benefit agricultural activities and contribute to local ecosystem services provision within their fencelines (Tan et al., 2025; Time et al., 2024), including carbon sequestration, pollinator supply, erosion reduction, and water retention (Ludzuweit et al., 2025). While agrivoltaics improve the ecological value of a site, an ecovoltaic approach aims to maximize ecosystem service provision within the GPV footprint by incorporating ecological principles in facility siting, design, and management (Sturchio and Knapp, 2023). Ecovoltaic interventions may overlap with those used in agrivoltaic systems (e.g., the planting of native pollinator-friendly plant communities), but typically include additional measures across GPV siting, design, and management decisions. Ecovoltaic facilities can be designed with corridors to maintain landscape connectivity (Chock et al., 2021), incorporate permeable fencing to allow wildlife movement into and out of the site (Buton et al., 2024), and include habitat features such as hedgerows, brush piles, or wetland patches to enhance the ecological value of the land within the fence line (Boscarino-Gaetano et al., 2024). These interventions are likely to be most potent for GPVs built on agricultural land, brownfields, or other heavily altered landscapes, where they may provide substantial local benefits to biodiversity (Tölgyesi et al., 2023). As such, it is possible for the area within a GPV footprint to retain some ecological function.

The discrepancies between datasets indicate that some are better suited to particular use cases. Increasing the availability of publicly accessible data to developers, policymakers, researchers, and other interested parties may, in part, help address the growing urgency of questions surrounding the impacts of an expanding US GPV footprint. The automated methodology of the Kruitwagen dataset is particularly appealing given that data on solar facilities, particularly for those in development or that have been recently constructed, are neither centralized nor easily accessible. Identifying the location of solar facilities from remote sensing imagery bypasses the need for manual review of the disparate county, state, and federal sources that maintain solar development data. Creating publicly accessible data from an automated pipeline also allows for further refinement of the process itself and broader utility of the data it produces. The Kruitwagen dataset seems to be especially adept at identifying and quantifying smaller arrays (≤ 1 MW) that may fall below the capacity thresholds used in datasets like the USPVDB, which includes facilities with capacities ≥ 1 MW, and the Wiki-

Solar Dataset, which typically includes facilities ≥ 3.5 MW. Thus, the prospect of automated pipelines capable of identifying GPVs as they are being developed is attractive. However, approximately 5 % of the Kruitwagen dataset inaccurately identified a GPV; the computer vision algorithm was confounded by mini-storage facilities, greenhouses, and other visually similar anthropogenic or natural features. As such, at this stage, automated machine learning pipelines for the delineation of US GPVs may be best used to identify the location of solar arrays prior to a human-assisted delineation of their footprints (Dai et al., 2024).

Of all the options explored here, the USPVDB is perhaps the most robust. The methodology it employs to digitize US GPVs is transparent, replicable, and reliable, creating a comprehensive publicly available database that lags only slightly behind the present solar landscape. Compared to the Kruitwagen dataset, the manual effort needed to establish and maintain this database is substantially higher. Populating this database using a more rigorous method, however, confers a higher level of accuracy (and therefore confidence) in the delineation of individual polygons. Caution should be used when applying this dataset to assessments of the broader facility footprint to the fenceline. Our results indicate that any analysis using data that only describes the cumulative array footprint can underestimate the cumulative facility footprint of US solar by approximately a third. Thus, the USPVDB is best suited for calculations of PV array land-use intensity of energy and analyses and those in need of the most current data.

Like the USPVDB dataset, the Wiki-Solar database relies on manual digitization, but lacks its transparency, replicability, and public accessibility behind a paywall. Unlike the other datasets we evaluated, Wiki-Solar data is not publicly accessible, which constrains future replication or extension of the footprint comparisons presented here. This runs counter to the growing emphasis on open science and FAIR (findable, accessible, interoperable, and reusable) data principles. This is unfortunate, given that our major conclusions regarding estimates of facility area and land-cover change rely on this dataset's unique delineation of GPV fencelines rather than panel arrays. As such, at the time of publication, our comparison of polygons in the Kruitwagen dataset and USPVDB represent the only accessible and replicable component of this analysis. We acknowledge the limitations this creates for transparency and reproducibility, even as its inclusion was central to this analysis.

Another limitation of the Wiki-Solar dataset is that the effort needed to digitize GPVs to their fencelines is more substantial than that of just the array footprint. Fencelines can be difficult to identify in remote sensing imagery, may contain irregular or convoluted segments, and are more likely to be obscured by vegetation. This may have also confounded our validation efforts, where visual checks were made by human individuals. Future work with this dataset or others like it could improve confidence in their analyses by validating the entire dataset and comparing accuracy ratings to those of an additional independent validation, such as the results of a machine learning classification. With that said, the Wiki-Solar dataset captures a meaningful portion of GPV-driven land-cover change that is not accounted for in the other datasets assessed here: those areas of a GPV that are typically subject to site preparation activities (e.g., blading, vegetation removal), operation activities (e.g. interior access roads), and other impacts to land beyond that which is covered by panel arrays. While this footprint does not capture all land-cover change driven by GPV development (e.g., road construction, transmission line corridors, vegetation management beyond the fenceline), the Wiki-Solar dataset does provide unique insight into the US GPV facility footprint. This dataset (and any other that represents the facility footprint) should be used primarily for assessments of the area occupied and impacted by GPVs, comparisons between the array footprint and facility footprint, and more nuanced calculations of GPV land-use efficiency at both the facility and array level.

Each of the datasets assessed here is useful given the right question, providing decision-makers with the information needed to guide these early stages of GPV expansion. The Kruitwagen dataset is excellent for

identifying as much solar as possible in a given landscape; the USPVDB is highly accurate and comprehensive for facilities 1 MW or greater; the Wiki-Solar dataset provides greater insight into land-related outcomes of GPV development. Their utility could be improved, however, with clear, standardized delineations of potential GPV footprints and minimum reporting standards that include publicly-available array and fence line extents for every US solar facility. Both are critical for scientific studies, where transparency, replicability, and comparability are essential. But they are equally important for policy and regulatory decisionmaking, where differences in standardized array and facility footprints could factor into permitting decisions or inform calculations of compensatory mitigation.

Establishing such standards will require navigating the complexity of other potential footprint delineations. For example, Lovering et al. (2022) use “direct area” to define the “land occupied by the electricity-producing facility” when calculating land use intensity of energy of major energy generation technologies (the area in hectares occupied by an energy producing facility per terawatt-hour of electricity generated in one year) This definition is similar to that of the facility footprint used in this analysis. However, the authors later clarify that the direct area of a GPV encompasses, “the area of panels or heliostats, roads established during development, and all ancillary facilities,” including features like transmission lines, switchyards, and meteorological towers, among others (Lovering et al., 2022). Standardized footprint delineations can reduce this type of ambiguity.

We recommend establishing these standards via a research-based workshop where a diverse group of experts discuss the criteria needed to demarcate GPV footprints at relevant spatial scales and consistently report them. Building on the USPVDB's array delineation methodology, the workshop could develop guidelines for digitizing facility fencelines, conducting quality assurance and control, and appending key facility attributes. These standards would be accompanied by guidelines for standard adoption and enforceability. This could include expanding reporting requirements to capture fencing length and materials (e.g., US Energy Information Administration Form 860, used by the USPVDB to collect GPV attribute information (Fujita et al., 2023)); incorporating the explicit submission of spatial data for facility and array footprints into local, state, or federal permitting processes; and creating a publicly accessible national dataset with standardized fenceline and array polygons for all US GPVs. By combining expert consensus with clear mechanisms for compliance, such a workshop would anchor footprint standards in both the scientific and policy realms.

Consider the potential impact of such a dataset on life cycle assessments (LCAs) for energy generation—comprehensive evaluations of the environmental impacts of energy generating systems throughout their life cycle, from the acquisition of their component parts to their ultimate decommissioning (Hauschild, 2019). Land-related considerations are often left out of such analyses for GPVs (van de Ven et al., 2021), in part because of a lack of quality data at relevant spatiotemporal scales (Jordaan et al., 2021). There are guidelines for the incorporation of land use into LCAs (de Baan et al., 2013; Koellner et al., 2013), though their inclusion has largely lagged behind that of other environmental considerations like greenhouse gas emissions (Jordaan et al., 2021); examples of land use as an LCA consideration can be found for fossil fuels (Jordaan et al., 2009, 2017), non-solar renewable energy technologies (Ciliberti et al., 2016; Dai et al., 2024), and concentrating solar energy [58]. More robust datasets capable of describing a GPV's array and facility footprint would be valuable in this context, particularly given some evidence that the incorporation of land use considerations into LCAs of PV solar can increase their estimates of lifetime carbon emissions by 10–150 % (van de Ven et al., 2021). There are even potential opportunities to consider the implications of innovation on future GPV land use requirements. For example, by mining data found in patent applications, patent-based prospective LCAs may inform forecasts of GPV land-use as more efficient, next-generation PV technology is developed (Spreatico et al., 2025). While the understanding of

relationships between land and energy remains nascent, these data are needed to provide researchers, industry, communities, decision-makers, and other interested parties with the robust information needed to guide GPV expansion in ways that may benefit people and nature.

CRedit authorship contribution statement

Michael O. Levin: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Emma Forester:** Writing – review & editing, Writing – original draft, Validation, Project administration, Methodology, Formal analysis. **Elizabeth L. Kalies:** Writing – review & editing, Supervision, Resources. **Lauren Goodman:** Validation. **Jason S. Hagani:** Validation. **Chevon C. Holmes:** Validation. **Noah Z. Krasner:** Validation. **Caitlin Markus:** Validation. **Jared B. Meek:** Validation. **Anubhav Vanamamalai:** Validation. **Shivani Agarwal:** Validation. **Uzma Ashraf:** Validation. **Daphne Condon:** Validation. **Deborah Forester:** Validation. **Lucy C.V. Holland:** Validation. **Elizabeth Jackson:** Validation. **Madeline Levin:** Validation. **Patrick F. McKenzie:** Validation. **Benjamin P. Narwold:** Validation. **Linnea E. Norton:** Validation. **Breanna Rodgers:** Validation. **Lucy C. Cheadle:** Validation. **David B. Dayan:** Validation. **Julia Dayan:** Validation. **Andrew Levin:** Validation. **Rebecca R. Hernandez:** Writing – review & editing, Supervision.

Data statement

All Google Earth Engine and R code used in this analysis has been documented and will be made publicly accessible via a Dryad repository upon confirmation the manuscript has been accepted. The Kruitwagen dataset is publicly available at <https://doi.org/10.5281/zenodo.5005867>. The USPVDB is publicly available at <https://doi.org/10.5066/P9IA3TUS>. We are unable to make the Wiki-Solar dataset public per the usage agreement for data purchased from Wiki-Solar. Both datasets are also available as Google Earth Engine assets, and the temporally harmonized dataset used to compare them in our analysis can be generated and exported with our Earth Engine code. The dataset is also included in the aforementioned Dryad repository. We are unable to make the Wiki-Solar dataset public per the usage agreement for data purchased from Wiki-Solar. Therefore, the harmonized dataset that will be accessible is limited to data from the USPVDB and Kruitwagen datasets.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Rebecca R. Hernandez reports financial support was provided by University of California Office of the President. Emma Forester reports financial support was provided by University of California Office of the President. Noah Z. Krasner reports financial support was provided by University of California Office of the President. Benjamin Narwold reports financial support was provided by University of California Office of the President. Chevon C. Holmes reports financial support was provided by National Science Foundation. Uzma Ashraf reports financial support was provided by Alfred P Sloan Foundation. Daphne Condon reports financial support was provided by Alfred P Sloan Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.127634>.

Data availability

Two of the three datasets examined here are publicly available. The third was purchased from the company Wiki-Solar, and the purchase terms state that the raw spatial data cannot be publicly shared.

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Glossary

- **Array Footprint** – the relatively contiguous area shaded by photovoltaic (PV) panels (i.e., shaded footprint), including the interspaces between the array areas.:
- **Facility Footprint** – the “array footprint” in addition to the other uncovered areas of the PV solar project within the facility’s fence line.:
- **USPVDB** – United States Photovoltaic Database.:
- **GPV** – Large, Ground Mounted Photovoltaic Solar Facility.: